# The Kelly Criterion for Optimal Bet Sizing

Antonio Melacini

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Consider a gamble with two states: we either win or lose.
Define the following variables:

- Let $W_0 > 0$ be your initial wealth.
- Let $x \in [0, 1]$ be the proportion of your wealth that you bet.
- Let $b > 0$ be the payout from a successful bet, so if you win you receive $W_0 bx$ and your total wealth becomes $W_0(1 + bx)$. If the odds are "$x$-to-$y$" then $b = x/y$.
- Let $a > 0$ be the proportion of your betting amount that you pay when you lose, so when you lose your wealth becomes $W_0(1 - ax)$. In an ordinary gamble $a = 1$ (you lose everything if you lose), but for a stock investment it might be reasonable to assume $a \in (0, 1)$. Also note that if you're leveraged (e.g., buying on margin), then $a > 1$.
- Let $p$ be the probability you win and $q = 1 - p$ be the probability you lose.

Now we make three key assumptions:

1. Bets compound.
2. Infinite repeated gambles are possible.
3. Gambles are independent.

Meaning that:

1. If we win once, we have $W_0(1 + bx)$, and then $100x\%$ of our wealth becomes $W_0(1 + bx)x$, so if we lose in the next bet, we will have $W_0(1 + bx)(1 - ax)$. This assumption holds up in most real-life settings.
2. We can continue playing the gamble repeatedly forever.
3. All rounds of gambling are independent. For example, whether we win or lose the next bet is not determined by whether we won or lost any of our past bets.

**Theorem (Kelly's Criterion):** Under the setup above, the optimal bet size is

$$x^* = p/a - q/b$$

and in the usual case where $a = 1$ we have Kelly's formula

$$x^* = p - q/b$$

**Proof:** Suppose we have bet for $n$ rounds. The number of rounds we won is a binomial random variable $N_{win} \sim \text{Bin}(n,p)$ and the number of rounds we lost is also a binomial random variable $N_{lose} = n - N_{win} \sim \text{Bin}(n,q)$. Hence, our wealth becomes

$$W_n = W_0(1+bx)^{N_{win}}(1-ax)^{N_{lose}}$$

so our wealth has multiplied by

$$R_n := \frac{W_n}{W_0} = (1+bx)^{N_{win}}(1-ax)^{N_{lose}}$$

Then, the per-period (or "per-bet") growth factor is:

$$R_n^{1/n} = (1+bx)^{N_{win}/n}(1-ax)^{N_{lose}/n}$$

Notice that $R_n^{1/n} = g(N_{win}/n, N_{lose}/n)$ where $g(u,v) = (1+bx)^u(1-ax)^v$ is a continuous function on the relevant domain here, and the Strong Law of Large Numbers guarantees that

$$\frac{N_{win}}{n} \xrightarrow[n\to\infty]{a.s.} p \quad \& \quad \frac{N_{lose}}{n} \xrightarrow[n\to\infty]{a.s.} q$$

so the continuous mapping theorem gives us

$$R_n^{1/n} \xrightarrow[n\to\infty]{a.s.} g(p,q) = (1+bx)^p(1-ax)^q =: R$$

So our asymptotic wealth accumulation rate per-round is $R$. We wish to find the optimal bet size $x^*$ that maximizes $R$;

$$x^* = \arg\max_x\{R\} = \arg\max_x\{\ln R\}$$

Let $L = \ln R = p\ln(1+bx) + q\ln(1-ax)$.

$$\frac{d}{dx}L = pb(1+bx)^{-1} - qa(1-ax)^{-1}$$

Then we solve for the root of the derivative:

$$\begin{aligned}
\frac{d}{dx}L = 0 &\implies 0 = pb(1+bx)^{-1} - qa(1-ax)^{-1} \\
&\implies pb(1-ax) = qa(1+bx) \\
&\implies pb - pbax = qa + qabx \\
&\implies xab(q+p) = pb - qa \\
&\overset{p+q=1}{\implies} xab = pb - qa \\
&\implies x = p/a - q/b
\end{aligned}$$

Therefore, the optimal bet size is

$$x^* = p/a - q/b \qquad (*)$$

When $a = 1$, the expression for $x^*$ simplifies to Kelly's formula $x^* = p - q/b$ $(*)$ can be verified to be the maximizer by calculating

$$\frac{d^2L}{dx^2} = -\left(\frac{pb^2}{(1+bx)^2} + \frac{qa^2}{(1-ax)^2}\right) < 0 \quad \forall x$$

Therefore $L$ is strictly concave and hence $x^*$ is a global maximum.

$\square$

What does our per-period wealth accumulation rate simplify to under this optimal betting?

$$\begin{aligned} R^* := R|_{x=x^*} &= \Big(1 + b(p/a - q/b)\Big)^p \Big(1 - a(p/a - q/b)\Big)^q \\ &= (1 + bp/a - q)^p (1 - p + aq/b)^q \end{aligned}$$

For example, say we play an all-or-nothing fair gamble ($a = b = 1$, $p = q = 1/2$). Then our optimal growth rate per-bet is

$$R^* - 1 = (1 + 0.5 - 0.5)^{0.5}(1 - 0.5 + 0.5)^{0.5} - 1 = (1)(1) - 1 = 0\%$$

which matches our intuition. What about when $a = b = 1$ but the house has an edge; $p = 0.49$ (and so $q = 0.51$)? In this case, $x^* = 0.49 - 0.51 = -2\%$, indicating that we should not play this gamble at all. In fact $x < 0$ is not a valid domain for our setup, so the optimal maximizer is just $x = 0$. This can also be interpreted as saying that the optimal strategy is to switch roles; be the house and deal this strategy rather than play it.